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Masked Language Modeling For Protein Sequence Generation

Abstract:
Protein binding is essential for almost all biological functions, including signaling, transport, conformational changes, and structural integrity. Innovations in peptide binding to create synthetic proteins and binding sites can foster new therapeutic solutions such as vaccines, as well as control and facilitate functions in biological contexts. Discovering new proteins via traditional wet lab methods is time consuming and expensive, and language models can bridge this gap by synthetically designing and testing many proteins simultaneously. The high specificity of these peptides can block other proteins from binding from a target protein or even act as inhibitors. Sequences can be easily generated via a model and tested with different metrics. Masked Language Modeling is a technique using a pretrained model that can predict a sequence of words, letters, or for our specific context, a biological sequence like an amino acid chain. BERT is an example of masked language modeling. In general, the model "masks" the middle of a specific sequence, and then uses the context on the left of that sequence as well as the context of the right of the sequence to predict what is missing in the middle. By doing this many times, the model will learn to predict sequences on its own. Masked language modeling also has various applications in other downstream Natural Language Processing (NLP) tasks, for example, used in sentiment analysis of social media posts, spam detection, named entity recognition, text completion, and can be finetuned for conversational analysis.

Literature Review:
Masked language modeling (MLM) is an unsupervised training approach that helps models learn how to understand sequences by predicting missing pieces of text. Before training, the text is broken down into tokens, which are the basic units the model processes. These tokens can represent full words, parts of words, or individual characters depending on the method used. During training, a small portion of these tokens are replaced with a special "[MASK]" symbol, and the model tries to predict the original content based on the surrounding context. Repeating this process across large datasets lets the model learn how tokens relate to one another and how meaning is formed within a sequence. As a result, the model develops a strong sense of context that can be applied to tasks such as classification, text generation, and sequence prediction, without requiring labeled data during the initial training phase.[9] For our literature review, we will be focusing on papers that use masked-language modeling in a health-related context.

In "DNABERT: Pre-trained Bidirectional Encoder Representations from Transformers model for DNA-language in genome," the authors used masked language modeling to better understand patterns within DNA sequence. Non-coding DNA is particularly difficult to study because its function depends heavily on the surrounding sequence context. Earlier models like convolutional neural networks and recurrent neural networks were able to pick up on some relevant patterns, but they placed too much of an emphasis on local regions and underperformed when modeling long-range relationships across entire sequences. They also relied on large amounts of labeled data, which is not always feasible in biological research. To address this, the authors adopted a transformer-based approach and used masked language modeling to learn directly from large amounts of unlabeled DNA.[5]

The researchers created the DNABERT model, which works by breaking down a DNA sequence into overlapping nucleotide sequences called k-mers and masking a portion of them during training. The model then attempts to predict the masked regions based on the rest of the DNA sequence. Since the model analyzes the entire sequence at once and picks up on regions that are most useful for prediction, it performs better than previous models that only looked at directly neighboring sequences. This model can be fine-tuned for specific tasks such as promoter prediction, transcription factor binding site identification, and splice site detection. DNABERT is especially useful to biological researchers because of its easy interpretability, allowing users to understand the model's predictions and their significance. Attention patterns highlight the regions of DNA the model is focusing on, so researchers can visualize which parts of the sequence are most important and often see that they correspond to known biological motifs. In many other machine learning models, the reasoning behind predictions is much less clear, which makes analysis more difficult. DNABERT also allows researchers to test how small changes in a sequence affect predictions, helping identify genetic variants that may have important functional or disease-related effects.[5]

When evaluated, DNABERT performed well across multiple tasks. For transcription factor binding site prediction, the model reached accuracy and F1 scores in the range of about 0.91–0.92. Precision and recall values were also high, indicating that the model identified true binding sites effectively while limiting false positives. Similar trends were seen in splice site prediction, with accuracy and F1 scores of around 0.92, along with a Matthews correlation coefficient (MCC) close to 0.87. In the case of promoter prediction, DNABERT showed higher accuracy, F1, and MCC compared to previous approaches. Since promoter datasets are often imbalanced, improvements in MCC indicate that the model is handling imbalance more effectively instead of favoring the majority class.[5]

In "FusOn-pLM: a fusion oncoprotein-specific language model via adjusted rate masking," the authors created a model to represent fusion oncoproteins. Fusion oncoproteins combine two genes through chromosomal rearrangements and are drivers of many cancers. However, they are difficult to target therapeutically due to high structural disorder and lack of stable binding pockets. Existing protein language models such as ESM-2 are trained on millions of natural protein sequences but do not explicitly include fusion proteins, which have distinct structural and functional properties. To address this gap, the authors develop FusOn-pLM, a fine-tuned protein language model specifically trained on fusion oncoprotein sequences.[12]

The proposed system builds on the ESM-2 architecture and introduces a new dataset and a modified masked language modeling (MLM) strategy. The authors constructed FusOn-DB, a curated dataset of over 44,000 fusion oncoprotein sequences. Then, they fine-tuned the final layers of ESM-2 using MLM using a cosine-scheduled masking strategy that dynamically varies the masking rate between 15% and 40% during training. This strategy increases task difficulty over time, encouraging the model to better capture unique sequence features of fusion proteins. The model was trained to reconstruct masked amino acids, and embeddings were extracted from the final layer for downstream tasks.[12]

The dataset used careful train-validation-test splits to reduce sequence similarity bias. Fusion proteins were shown to be structurally distinct from normal proteins, with significantly higher intrinsic disorder (about 45.9% disordered residues compared to ~33% in their component proteins). Multiple model configurations were evaluated, including fixed masking rates and various dynamic masking strategies. The cosine-scheduled masking approach performed best, achieving a loss of 1.28 and pseudo-perplexity of 3.61, outperforming both the baseline ESM-2 model and other masking strategies. Pseudo-perplexity is an approximate metric for perplexity that measures how well a masked language model predicts each token in a sequence using full context, with lower values indicating better performance.[12]

To evaluate the model, the authors tested FusOn-pLM on several biologically relevant downstream tasks. First, they evaluated puncta formation and localization prediction using XGBoost classifiers trained on model embeddings. FusOn-pLM outperformed baseline embeddings (ESM-2, ProtT5, and handcrafted features) across multiple classification metrics. Second, the model was used to predict properties of intrinsically disordered regions (IDRs), such as radius of gyration and scaling behavior, achieving strong correlations between predicted and true values. Lastly, the model performed zero-shot mutation prediction. By systematically masking individual residues, FusOn-pLM predicted likely amino acid substitutions and identified functionally important or conserved regions. The model successfully identified known drug-resistance mutations in fusion oncoproteins. It was able to identify resistance-associated mutations in 12 out of 14 known sites in EML4::ALK and in 13 out of 28 sites in BCR::ABL. These evaluation tasks signified that FusOn-pLM was successful in modeling fusion oncoproteins and capturing biologically relevant information.[12]

In "Focused learning by antibody language models using preferential masking of non-templated regions," the authors improved standard MLM strategies for antibody-specific protein language models. Traditional MLM applies a uniform masking rate across all positions in a sequence. However, antibody sequences are highly uneven in their structure, with certain regions playing a much more important role in antigen binding due to their high variability and functional significance. One such region is the complementarity-determining region 3 (CDR3). The authors argued that uniform masking does not adequately prioritize these critical regions, limiting the model's ability to learn meaningful biological patterns. To address this, they introduce a preferential masking strategy that increases the masking rate in the CDR3 while maintaining the same overall masking proportion. (Ng & Briney, 2025).

The proposed system modifies the masking phase of MLM while keeping the overall model architecture consistent. Instead of masking tokens uniformly, the model applies a higher masking probability (25%) to residues in the CDR3, ensuring that the model spends more time learning to predict tokens in the most variable and functionally important regions. The model is trained using an ESM-2 transformer architecture with approximately 350 million parameters, and antibody sequences are represented as concatenated heavy and light chains.[10]

The dataset used for training consisted of approximately 1.6 million paired antibody sequences collected from healthy human donors, along with several additional datasets designed for downstream evaluation tasks. These include datasets for native versus shuffled heavy and light chain pairing, as well as antigen binding specificity classification. Using this data, the authors compared the baseline model trained with uniform masking and a preferential masking model that focuses more heavily on the CDR3 region. Overall, the preferential masking model consistently performed better across multiple evaluations. It achieved similar or improved validation performance while requiring about 29% fewer training steps, demonstrating greater training efficiency.[10]

In downstream tasks, the preferential model showed stronger predictive performance. The preferential model was able to predict antibody chain pairing at a higher accuracy than the baseline, reaching around 0.66 on one dataset and 0.62 on a more challenging dataset. Therefore, the model was better able to capture non-random patterns that influence antibody chain pairing. The preferential masking model also performed better in antigen binding specificity classification, particularly for SARS-CoV-2 antibodies, indicating an improved ability to learn biologically meaningful sequence features. To further understand how the model makes predictions, the authors applied an explainable AI method called Attentive Class Activation Tokens (AttCAT). This analysis showed that the model primarily relies on residues within the CDR regions (especially the heavy-chain CDR3) when determining binding specificity, reinforcing the importance of these regions in antibody function.[10]

In "Generating Synthetic Free-text Medical Records with Low Re-identification Risk using Masked Language Modeling", the authors explore a different approach to utilizing masked language modeling (MLM). Instead of using the MLM to deal with incomplete data, the authors start with complete patient records and intentionally obscure parts of them through reconstruction. Patient data is extremely valuable for machine learning, but its use is limited due to privacy concerns. To address this, the authors use MLM to remove identifying information while keeping important medical content intact. Prior approaches have included causal language modeling (CLM), which predicts the next token in a sequence. However, CLM is unidirectional and does not consider the full context of the data. While it maintains data integrity, it does not sufficiently protect patient privacy. Another approach has been to generate entirely synthetic medical records using GANs, but these often lose important medical details, making them less useful for training.[1]

The proposed system consists of two main components: a masking phase and a reconstruction phase. In the masking phase, identifying information is first removed using regex rules targeting categories such as date, ID, name, contact, age, and location. Then, named entity recognition (NER) is applied to ensure that key medical information such as the problem, test, and treatment, is preserved. Additionally, part-of-speech tagging is used to randomly mask other words, with control over the percentage of nouns, verbs, and adjectives that are masked. In the reconstruction phase, a pre-trained MLM called Bio_ClinicalBERT (trained on clinical data) is used to fill in the masked tokens. Two different strategies are explored for this step. The first is simultaneous chunk filling, where all masked tokens are filled at once. The second is iterative mask filling, where tokens are filled one at a time, with each newly generated token influencing the next prediction.[1]

The dataset used in this study consists of records from diabetic patients. The system is optimized using several parameters, including the MLM learning rate, training batch size, the proportion of protected health information (PHI) masked, and the overall masking probability. Four model variations were tested: stochastic masking of 50% and 70% of nouns, verbs, and adjectives, and iterative masking at 70% and 90%. Additionally, two synthetic samples were generated from each real data point, effectively increasing the dataset size. For evaluation, the authors used an existing model called SciSpacy to extract disease and chemical entities from electronic health records. They compared performance when training on original data versus masked data. The results showed that the original data achieved an F1 score of 0.842, while the best-performing masked model (iterative 90%) achieved a very similar score of 0.836.[1]

In terms of privacy, the masking system performed well. It achieved a recall of 96% in identifying PHI (based on manually annotated datasets), and only 3.5% of that PHI could be re-identified after masking. This indicates that the method is effective at protecting sensitive information. This study was important because it opens up many more possibilities for the utilization of real medical data to improve machine learning models in an ethical way that protects patient privacy.[1]

Discussion:

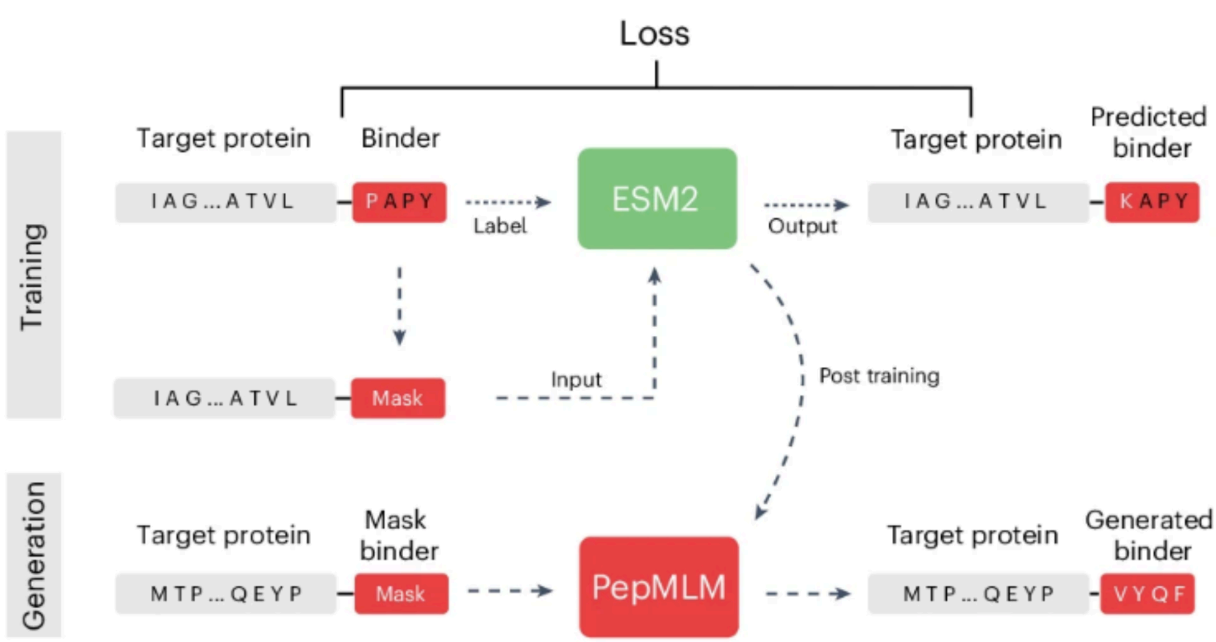
PepMLM, short for Peptide binder design algorithm via Masked Language Modeling, is a new approach that applies traditional context-dependent masked language modeling to antibody sequences of amino acids. The project was led by Pranam Chatterjee, an assistant professor at the University of Pennsylvania. Leo Tianlai Chen from Duke University led the core development of PepMLM by designing its architecture, curating the peptide protein datasets, and training and evaluating the models, as well as carrying out benchmarking with support from collaborators. The broader experimental validation and analysis were distributed across team members from various American institutions, with Zachary Quinn (Duke University) handling binding assays, Christina Peng (Cornell University) conducting immunofluorescence and degradation assays, and other researchers contributing to plasmid construction, viral studies, and testing the resulting peptides. Overall, Pranam Chatterjee developed the idea and directed the entire project, coordinating both the computational design and biological validation efforts. PepMLM designs de novo, or novel, antibodies not typically available in wet lab contexts, and can be used for applications such as drug design or vaccine development. PepMLM is grounded in biochemistry because it designs peptide sequences whose amino acid composition determines intermolecular interactions such as hydrogen bonding, electrostatic interactions, and hydrophobic effects. It is also closely tied to biology, since these generated peptides can function as binders to proteins, influencing biological pathways relevant to drug discovery, immunology, and disease treatment. By connecting sequence patterns to functional binding behavior, the model bridges biochemical principles with the computational design of new biomolecules.[2]

PepMLM's primary purpose is to enable fast, structure-free design of peptide therapeutics, particularly for conformationally challenging targets that lack well-defined 3D structures, which is why it is trained only on sequential data. PepMLM generates linear peptides and considers only the amino acid sequence, not structural information, meaning it does not require Protein Data Bank files. How these peptides fold after generation is evaluated using an external model such as AlphaFold. Here, we discuss the model architecture of PepMLM and describe how it generates peptides from an initial target sequence, which is typically around 500 amino acids long. The generated binder length can range from 15–50 amino acids. All the data used to train PepMLM in the paper is publicly available, allowing reproduction of the reported results. The dataset was curated from protein–peptide complexes obtained from the PepNN (Peptide-protein Neural Network) and Propedia databases. Because the datasets were somewhat similar, duplicates were removed to reduce redundancy. The researchers then used MMseqs2 (Many-against-Many sequence searching), a tool for clustering large protein sequence sets, applying an 80% similarity threshold.[2] Redundancy within each cluster was further reduced, followed by manual removal of highly similar entries. The final dataset consisted of 10,203 sequences, split into approximately 10,000 for training and about 200 for testing. This differs from the traditional 80–20 split, likely because masked language modeling benefits from maximizing training data, especially given the relatively small dataset. A smaller test set is more acceptable in this context, particularly when combined with additional validation methods beyond computational metrics. We now present the model architecture of PepMLM.

Model Architecture:

The de novo peptide binder design tool fine-tunes the ESM (Evolutionary Scale Modeling)-2 protein language model using a span masking strategy. ESM-2 was originally pretrained on hundreds of millions of protein sequences to learn evolutionary relationships between amino acids and proteins. During training, PepMLM masks the entire peptide binder sequence and places it at the C-terminus of the target protein sequence.[2] This forces the fine-tuned ESM-2 model to reconstruct the masked region using the context provided by the target sequence alone. This is the core idea behind the training method. ESM-2 is an encoder-only transformer model, which means that it specializes in learning bidirectional contexts of sequences rather than generating sequences directly.[8] By fine-tuning it on known peptide–protein pairs, the researchers train the model to effectively "fill in the blank" for what a good binder should look like in a specific target. The model is trained using cross-entropy loss, which measures the performance of classifying models by calculating the logarithmic difference between predicted probability and actual one-hot encoded labels in distributions. This loss function guides the model during training by adjusting parameters to minimize prediction error. To generate a certain protein, the model accepts a target sequence and a user-defined number of mask tokens corresponding to the desired peptide length, and reconstructs the binder sequence to fill those positions.[11] Two generation strategies were used. The first was a greedy approach in which the model looks at all possible amino acids and their predicted probabilities and picks the single most likely one. The second was a top-k sampling strategy, where the model considers the top three most likely amino acids (k=3), randomly samples one based on their probabilities, balancing sequence diversity while maintaining model confidence.[2]

Figure 1.



In Figure 1 above, the PepMLM model is separated into 2 phases: the training phase and the generation phase. In the training phase, the input given is the target protein and the masked binder, and the actual binder sequence is used as a label for the model. Then, the ESM2 model produces an output of the predicted sequence of the binder with the target protein. From there, the loss is calculated and constantly updated for each position in the binder.

The loss was calculated based on a peptide $p = (p_1, p_2, ..., p_n)$ of size n, and a binder $b = (b_1, b_2, ..., b_m)$ of size m. Masked language modeling tries to recreate the binder b by finding the probability of the amino acid in the specific position of the binder given the peptide and the masked binder. This loss is calculated as:

$$L_{MLM} = -\frac{1}{m}\sum_{i\in m} log\, Prob(b_i | p,\, b_{mask})$$

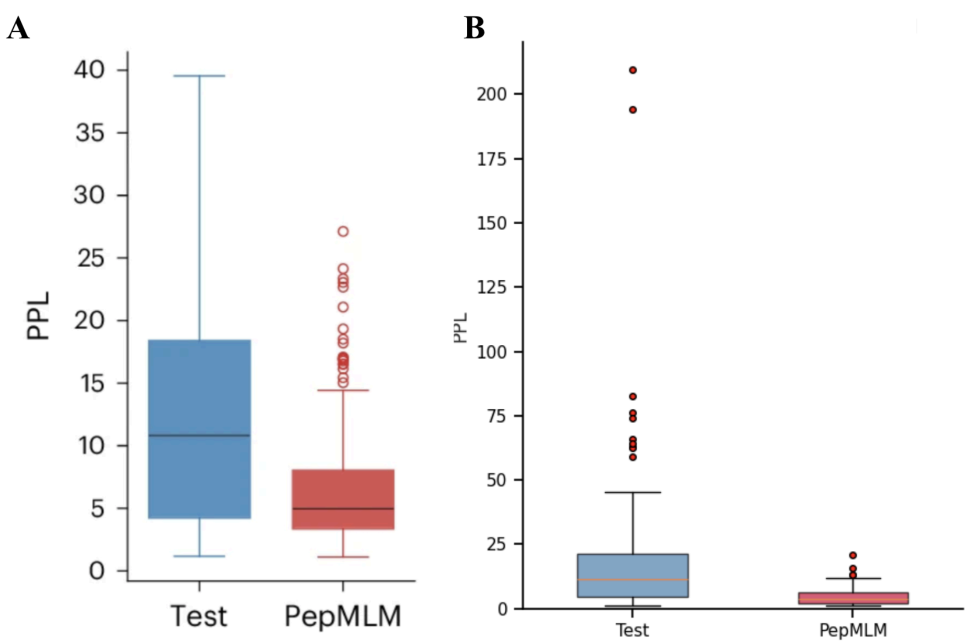
By minimizing the loss, the difference between the predicted and actual binders is also minimal, which reduces the problem to the approximation $\prod_{i=1}^{m} P(b_i \mid p)$.

The primary evaluation metric that the authors used was pseudo-perplexity (PPL). Lower pseudo-perplexity values indicate higher sequence fitness, meaning the sequence is more likely and more consistent with patterns learned during training. Mathematically, it is calculated using the exponentiated average negative log-likelihood assigned to each token in the sequence, denoted as:

$$PPL(b) = exp\{-\frac{1}{m}\sum_{i=1}^{m} log\, Prob(b_i \mid b_{j\neq i},\, p)\}$$

In this study, pseudo-perplexity was calculated by scoring each amino acid position in the binder one at a time while masking it, and averaging the resulting scores from all the positions. Pseudo-perplexity is an approximate metric for perplexity that measures how well a masked language model predicts each token in a sequence using full context, with lower values indicating better performance.[2] Below are the PPL values after running the model as reported in the paper (Figure 2A) and the PPL values after our own test case of the model (Figure 2B).

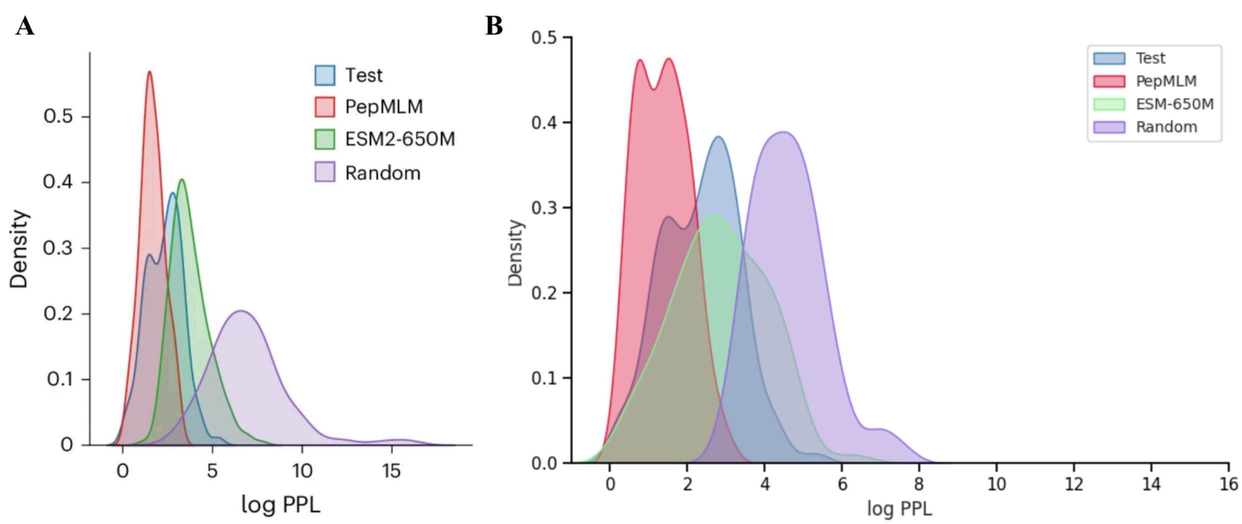
Figure 2.



The purpose of the model run depicted in Figure 2A is to directly test whether PepMLM-generated peptides achieve perplexity scores comparable to experimentally validated binders from the test set. We can clearly see that PepMLM exhibits a lower perplexity value than the test cases as well as a narrower range. Less variability in PPL could indicate that the model is consistent and stable, which are characteristics of a successful predictive model. These same trends were also seen in our test run, as shown in Figure 2B. This figure proves that the peptides generated by PepMLM have biological significance and value, and we were able to recreate these trends using our model.

Continuing the analysis of perplexity, Figure 3B examines this metric and compares the four different groups of real peptides, PepMLM-generated peptides, ESM-2 output peptides, and random sequences of peptides. This is a density plot, so it uses normalized counts. The benefits of using density are that the visuals are independent of sample size, which lets us compare fairly across groups. Furthermore, density plots make it easier to see shifts in curves. When the curve shifts towards the left, it indicates lower perplexity. Conversely, when the curve shifts towards the right, higher perplexity is indicated. The results from the paper are shown in Figure 3A, whereas the results outputted by our model are shown in Figure 3B.

Figure 3.

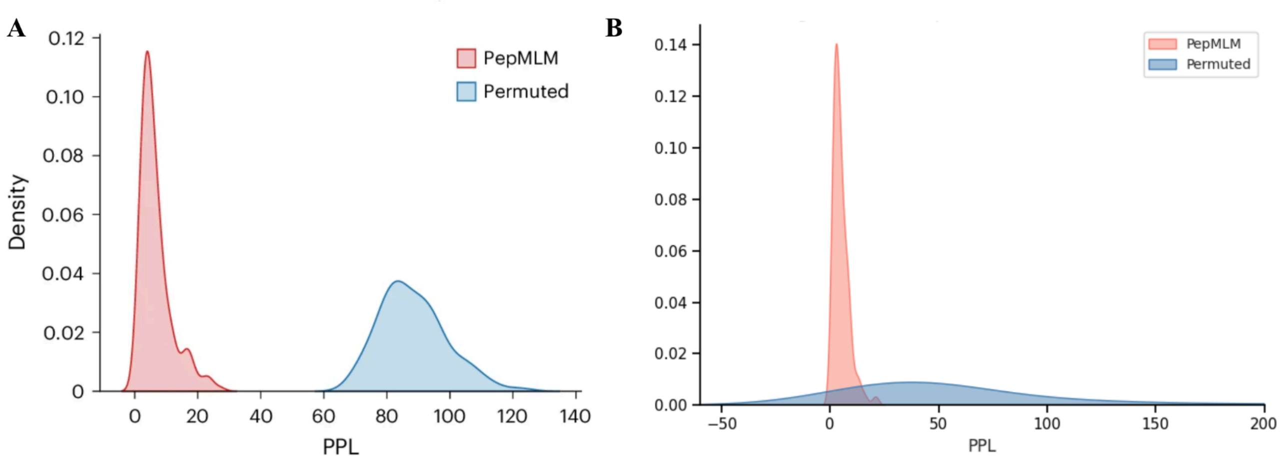


The distribution analysis from Figure 3, for both the paper results and our comparative results, reveal that PepMLM closely mirrors the low perplexity region of real binders. This is a stark difference from the distribution observed with the original ESM-2 model alone and with randomly designed binders, indicating that PepMLM can distinguish binders from non-binders through PPL scores. PepMLM-generated peptides are biologically plausible, while ESM-2 alone (without the specialized binder fine-tuning) doesn't capture binder-specific patterns well and random sequences produce distributions shifted toward higher perplexity, confirming that the fine-tuning done by PepMLM was essential and meaningful. Our model test run observed similar results when comparing the ESM-2 model. The PepMLM curve, denoted in red, has a lower density for the peak as reported in the paper, but it can still be observed to be shifted to the left when compared to the other two groups.

Aside from pseudo-perplexity, the paper uses the interface-predicted Template Modeling ipTM score from AlphaFold-Multimer as a computational metric. AlphaFold-Multimer is an extension of AlphaFold2 designed to predict the 3D structure of protein complexes rather than single chains.[2] The metric ranges from 0 - 1 and exists to assess the accuracy of predicted protein-protein interaction interfaces. Scores above 0.8 indicate high confidence and scores below 0.6 indicate failed generation. Scores between 0.6 and 0.8 fall within a grey zone left for interpretation. Here, we use ipTM at the binding interface, specifically focusing on the confidence of the predicted interaction between two chains at the spot they bind at. This strategy makes ipTM a direct way to test predicted binding quality, which is another important feature of generating peptides. The authors also make use of pLDDT(predicted Local Distance Difference Test) and AlphaFold's per-residue confidence score, which measures how accurately the model thinks it has predicted the position of each atom. High pLDDT values at a binding spot show us that predicted binding interactions are valid, while low high pLDDT scores are associated with lower prediction accuracy. To validate the consistency of the PPL metric, the authors showed that it correlates with folding scores and with extracted ipTM and pLDDT values from benchmarking. A statistically significant ($p < 0.01$) negative correlation was seen with PPL, demonstrating that the model can reflect binding interactions in a zero shot manner while validating its reliability in prioritizing stable target-binding molecules. This negative correlation means that as PPL goes down, or as the model is more confident in the peptide, the ipTM score goes up. Therefore, both metrics converge on the same answer.

The discussions on perplexity earlier failed to account for a significant issue. While it is important that sequences are biologically plausible, perplexity does not recognize that they must also exhibit good affinity for a certain target sequence. To help examine this case, the authors of the paper ran a permutation analysis. They took the target binder pairs and then scrambled them so they were with different targets. Correct pairs should report a lower PPL and the permuted pairs should report a higher PPL. Figure 4 tests the integrity of PPL as a valid metric. The results generated by the paper are shown in Figure 4A, whereas the outputs from our test case are shown in Figure 4B.

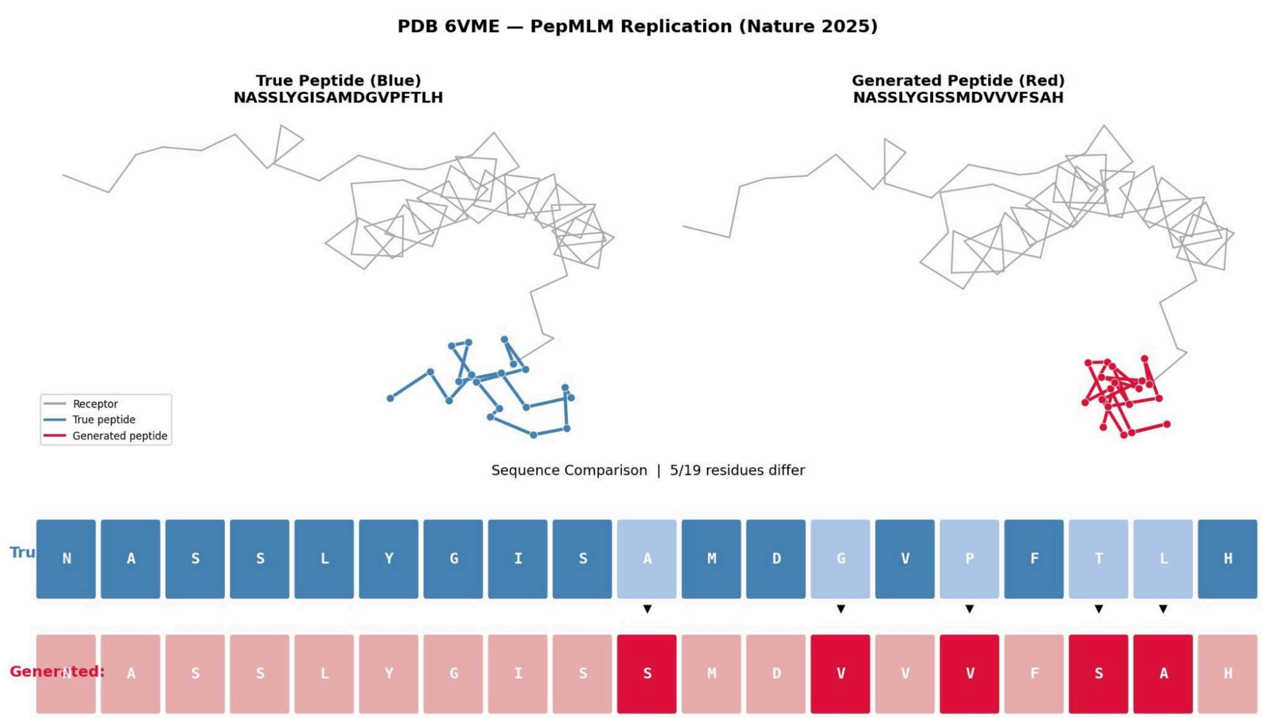
Figure 4.



For the sake of computational efficiency, we set the number of random binder shuffles to ten. However, for each target in the paper, one hundred random binder shuffles were performed and PPL scores were computed for those mismatched pairs. Statistical analysis using t-tests revealed significant differences ($p < 0.001$) between the PPL distributions of matched and mismatched pairs, with consistently higher PPL values being observed in shuffled pairs. These results indicate that designed binders exhibit target specificity, as disrupting the intended target binder pairing leads to predicted lower binding affinities.[2] Thus, even though all the sequences are chemically reasonable peptides, the model gives them higher perplexity when paired with the wrong target, demonstrating that the PPL scoring metric tests for genuine target specificity and generic protein-like properties.

The results outputted in Figure 4B present a replication and study of PepMLM's generative behavior using PDB entry 6VME. This PDB file was generated by ESMFold, which is also an extension of the ESM-2 model from Meta AI that generates the structure from a given peptide sequence. The model was run on a sample of sequences drawn from the training data to generate candidate peptide binders and compare them directly to their experimentally validated test counterparts. In this particular example, the PepMLM generated peptide achieved a lower PPL than the test peptide. This suggests that in some cases, PepMLM is not merely approximating the known binder but is in fact finding an alternative sequence that it considers more compatible with the target. At the sequence level, the comparison is shown in the bottom panel, where the true peptide (NASSLYGISAMDGVPFTLH, blue) and the generated peptide (NASSLYGISSMDVVVFSAH, red) are aligned side by side with arrows marking the positions where the sequences are different. Out of 19 total amino acid residues, only 5 differ, noted at positions 10, 13, 16, 17, and 18. These correspond to the amino acid substitutions Alanine to Serine, Glycine to Valine, Proline to Valine, Threonine to Serine, Leucine to Alanine. The first nine residues (NASSLYGIS) are shared between both sequences, so PepMLM was most likely able to correctly learn the N-terminal portion of the binder because this portion was close to the target binding interface as there is more information present. Not all the swaps were notable. For instance, threonine to serine is not a relevant swap because both amino acids are small polar hydroxyl having residues. However, the swap from proline to valine is notable, since proline is helix-breaking and cyclic and valine is not, which will probably alter the local backbone geometry of the peptide at that position.

Figure 5.



The most useful aspect of this figure, however, is the structural comparison shown in the backbone trace diagrams above. Both panels display the same gray receptor structure, with the true peptide shown in blue (left) and the PepMLM generated peptide in red (right). Despite the high sequence similarity at the generated regions, the two peptides adopt completely different conformations. The true peptide (blue) adopts a relatively extended, loose structure with no tight compact clustering. In contrast, the generated peptide (red) folds back onto itself into a much more compact, densely packed globular cluster, with tight turns and low volume. This difference reflects different dihedral angles, different secondary structure, and potentially a change in binding. The fact that the generated peptide achieves a lower PPL despite folding so differently suggests that PepMLM is not simply learning to mimic the exact conformation or sequence of known binders, but is instead exploring alternative sequences that satisfy the target's chemical compatibility requirements through a structurally distinct solution without taking in any structural data.

Conclusion:

In conclusion, masked language modeling (MLM) is useful in chemical and biological situations because it can learn the grammar of molecular sequences like amino acid chains from purely sequential data and then predict missing parts based on context, capturing underlying chemical relationships without explicit rules. This allows models to generate new molecules or peptides that preserve key biochemical properties such as binding affinity and stability, which are critical for drug design and vaccine development. However, beyond just proteins, masked language modeling can be applied to other sequence-like chemical representations, especially where we can follow the sequence to structure pipeline.[9] For example, it can help design small molecules using representations like SMILES strings, enabling drug discovery by generating novel compounds with desired properties. It can also be used in materials science to explore polymer sequences or inorganic compositions, where learned patterns help predict stability, conductivity, or reactivity. Ultimately the work flow is sequential data fed into the model, which recognizes patterns, and outputs sequential output, which can be filtered and tested for structural implications. However, an additional implication of this field is wondering how structural data can be directly integrated into the generation process itself. This would allow the model to learn not just sequence patterns but also the spatial constraints which govern folding and binding. Even the environment of where a peptide is typically found, or would be placed, could be a valuable piece of information. For example, a peptide in the stomach would have to deal with different environmental pH constraints than a peptide found in the pancreas. This could improve the physical realism of generated molecules by coupling sequence design with explicit representations of 3D structure during training. This is an idea that we are interested in looking into in the future, as well as the idea of inverse folding. Inverse folding means that the process is reversed. Instead of sequential data, we give a model structural data, a fixed 3D backbone structure as input and it outputs amino acid sequences which will fold into that structure.[7] There are many areas where this work using MLMs can expand into, and integrated into various workflows, regardless of whether they are chemical contexts or not. It is a new and exciting field that is constantly changing. By working directly from sequence data, MLM enables discovery of chemical spaces we didn’t even know existed.

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